

Comic Image Inpainting via Distance Transform

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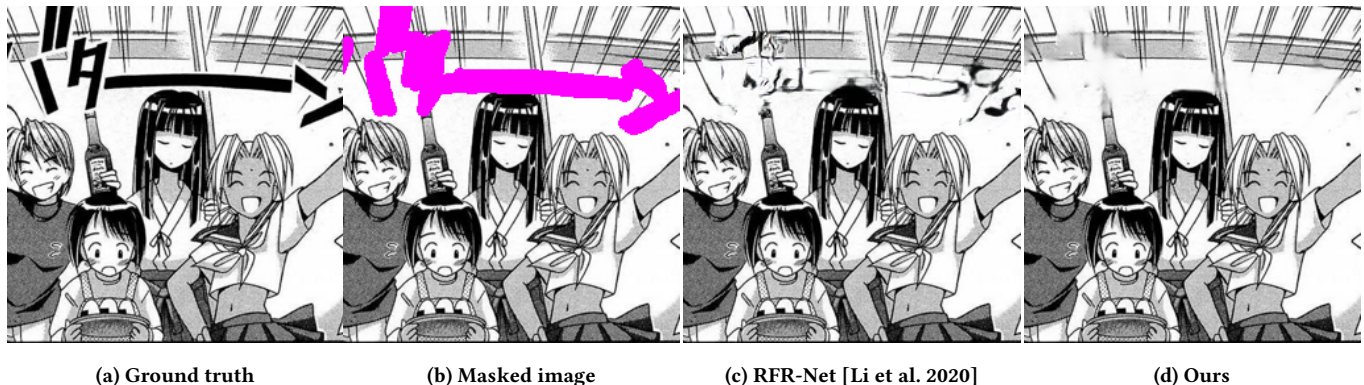


Figure 1: An example of onomatopoeia removal using inpainting. The left image (a) is the ground truth (LoveHina ©Akamatsu Ken). The goal is to complete the missing region given in purple in the center image (b). The right two (c)(d) are the inpainted results. Because the mask covers the onomatopoeia, our method removed the onomatopoeia in the restored image.

ABSTRACT

Inpainting techniques for natural images have progressed significantly. However, if these methods are applied to comic images, the results are not satisfactory because of very noticeable artifacts, especially around line drawings. Line drawings are challenging to inpaint because of their high-frequency components. In this paper, we propose a novel method for inpainting comic images in the distance transform domain. In this method, we first convert a line drawing into a distance image and then inpaint the distance image. By transforming line drawings into distance images, we can eventually reduce the high-frequency components, which leads to improve inpainting performance. We compared the results of our proposed method with those of the conventional methods. The results showed that the proposed method achieved 0.1% lower l_1 loss, 0.5dB higher PSNR, and 0.5% higher SSIM than those of the conventional inpainting methods.

CCS CONCEPTS

• Computing methodologies → Image processing.

KEYWORDS

inpainting, comic, neural networks

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1 INTRODUCTION

Inpainting is a task for completing the missing regions of an image. In this paper, we focus on inpainting of a comic image. Inpainting a comic image enables us to perform advanced comic image processing. An example is erasing onomatopoeia (Figure 1) and repositioning the translated one. Such an operation is important to achieve a fully automated comic translation [Hinami et al. 2021].

Although many researchers have proposed inpainting methods for natural images [Iizuka et al. 2017; Li et al. 2020; Nazeri et al. 2019; Yu et al. 2018], it is not easy to apply them to comic images. The reason is that inpainting of high-frequency components is generally difficult. Therefore, when line drawings with high-frequency components are inpainted, the results often have artifacts or lines broken in the middle, as shown in Figure 1(c).

In this paper, we propose an inpainting method that converts a line drawing into a distance image [Rosenfeld and Pfaltz 1966] and inpaints the resulting distance image. On the assumption that the original comic image is binary, each pixel of the distance image represents the nearest distance to the black pixel in the original binary image. A distance image obtained from a line drawing contains more low-frequency components than a line drawing because each pixel of the distance image has a continuous value. Therefore,

we can improve the accuracy by inpainting the distance image and converting the result into a line drawing.

We trained the proposed model on the comic image dataset Manga109 [Aizawa et al. 2020; Matsui et al. 2017] and compared it with other methods for natural images. The results quantitatively showed that the proposed method outperformed the existing methods. By comparing the inpainted results, we confirmed that the proposed method has an advantage in line inpainting.

2 RELATED WORK

In this section, we first describe the inpainting methods for natural images. Next, we explain the distance images and the related works that used them.

2.1 Natural image inpainting

Many studies have developed the inpainting methods for natural images for a long time. Since the advent of deep learning, methods using generative adversarial networks (GANs) [Goodfellow et al. 2014] have become the mainstream. Iizuka et al. used two networks, namely, a local discriminator and a global discriminator [Iizuka et al. 2017], which helped to ensure global consistency and local sharpness in the results. Nazeri et al. [Nazeri et al. 2019] proposed EdgeConnect, a method for generating more acceptable images by inpainting only the edge components first. Li et al. proposed RFR-Net [Li et al. 2020], which successfully inpaints a wide range of mask regions by performing the inpainting in stages. Recently, Xie et al. proposed an inpainting method for comic images [Xie et al. 2021], which is contemporaneous work with our work. We will discuss a comparison with this method in Section 4.2.

2.2 Distance transform

A distance transform [Rosenfeld and Pfaltz 1966] is a type of image created from a binary image. The value of each pixel in the distance image represents the minimum distance from any black point in the original binary image. Distance images are helpful for filling in spatially sparse information, such as line drawings in cartoon images. Narita et al. transformed line drawings into distance images when estimating the optical flow of line drawings [Narita et al. 2019]. Almatrafi et al. transform instantaneous input information into a tractable form by applying a distance transformed to the luminance changes obtained from an event camera [Almatrafi et al. 2020].

3 METHOD

In this paper, we propose a novel method for inpainting comic images. Figure 2 shows the overall diagram of the process. Inpainting of line drawings is difficult because they contain high-frequency components. Therefore, we propose a method that emphasizes the inpainting of line drawings.

Let us denote an original (groundtruth) comic image as $I_0 \in \mathbb{R}^{H \times W \times 3}$ and a mask as $M \in \{0, 1\}^{H \times W \times 1}$. Here, H and W are the height and the width of the image, respectively. We assume that a comic image is a three-channel real-valued tensor. A mask is a single-channel tensor with a value of 0 or 1, whereas "1" indicates the mask (orange region in Figure 2). Our system receives a missing comic image I_m , i.e., the image I_0 with some pixels covered by the mask M . The system then outputs an inpainted comic image

$\tilde{I} \in \mathbb{R}^{H \times W \times 3}$. Our goal is to recover the original image as much as possible: $I_0 \sim \tilde{I}$.

In the proposed method, the inpainting of a line drawing consists of two tasks: (1) inpainting of the distance image and (2) conversion of the distance image into a line drawing. By converting the line drawing to a distance image once, we can reduce the frequency of the image. Therefore, the difficult task of inpainting a line drawing can be reduced to a much easier task. Finally, we refer to the structure of EdgeConnect [Nazeri et al. 2019] and perform (3) screentone inpainting using the restored line drawing as an auxiliary. A GAN composes the module for each inpainting. The process represented by the dotted line in Figure 2 is used only for the training.

3.1 Preprocessing

In this section, we describe the preprocessing in the training phase. We take the original comic image I_0 and the mask M as inputs and output the following five images: the original line drawing $U_0 \in \mathbb{R}^{H \times W \times 1}$, the original distance image $V_0 \in \mathbb{R}^{H \times W \times 1}$, the missing comic image $I_m \in \mathbb{R}^{H \times W \times 3}$, the missing line drawing $U_m \in \mathbb{R}^{H \times W \times 1}$, and the missing distance image $V_m \in \mathbb{R}^{H \times W \times 1}$. Here, a line drawing U_0 is an image that consists only of the lines extracted from the original comic image I_0 , without textures such as tone or black fill. In the test phase, we prepare the missing comic image I_m , the missing line drawing U_m , and the mask M .

Figure 3 shows the overall diagram of the preprocessing. First, we extract the original line drawing U_0 from the original comic image I_0 using Li et al.'s method [Li et al. 2017]. Next, we generate the original distance image V_0 from the original line drawing U_0 . Second, we apply the distance transformation f to the original line drawing U_0 and obtain the original distance image $V_0 = f(U_0)$.

Next, we describe the procedure for obtaining the missing comic image I_m , line drawing U_m , and distance image V_m . First, we obtain the missing comic image I_m by multiplying the original comic image I_0 and the mask M . Similarly, we generate the missing line drawing U_m by using the original line drawing U_0 and the mask. The missing distance image is obtained by converting the missing line drawing U_m into a distance image $V_m = f(U_m)$, as shown in Figure 3.

3.2 Inpainting of distance image

Here, we describe the details of each module. Let us define the network that restores the distance image as $G_{\text{dist}} : \mathbb{R}^{H \times W \times 1} \times \{0, 1\}^{H \times W \times 1} \rightarrow \mathbb{R}^{H \times W \times 1}$. The network receives the missing distance image V_m and the mask image M as inputs and returns the inpainted distance image $\tilde{V}$. To train the network to restore the distance image correctly in the mask region, we introduce $\mathcal{L}_{l_1}$, the normalized l_1 distances between the inpainted distance image $\tilde{V}$ and the ground truth distance image outside the mask region $V_0 \odot (1 - M)$. Similarly, $\mathcal{L}_{l_1 \text{mask}}$ is defined as the normalized l_1 distances inside the mask M . We also add the adversarial loss $\mathcal{L}_{\text{adv}}$, which is used in GAN training, to obtain a clearer image. Furthermore, to match the structure and style of the image, we introduce the perceptual loss $\mathcal{L}_{\text{perc}}$ and style loss $\mathcal{L}_{\text{style}}$ proposed in [Gatys et al. 2016; Johnson et al. 2016]. Lastly, the loss function of G_{dist} is defined as follows:

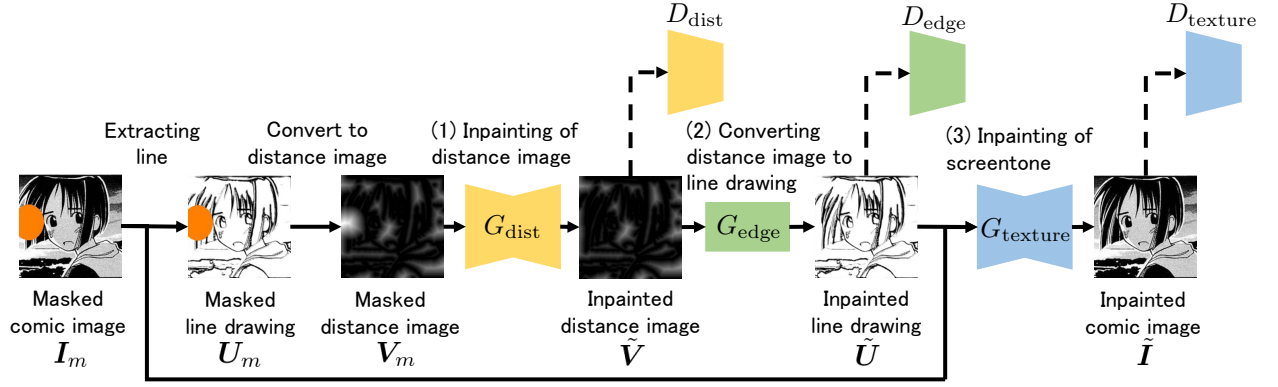


Figure 2: Proposed method (LoveHina ©Akamatsu Ken). The inputs to the discriminators D_{dist} , D_{edge} , and D_{texture} (dotted line) are provided only during training.

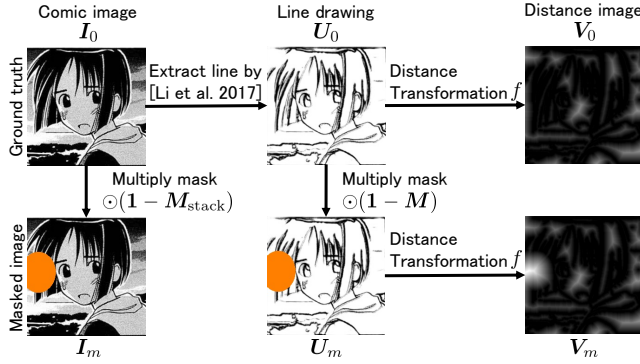


Figure 3: Generation of the input data. The image is from the Manga109 dataset (LoveHina ©Akamatsu Ken). Note that $\odot$ denoted an element-wise multiplication, M_{stack} is a stacked M over the channel direction, and 1 indicates the all-one tensor.

$$\begin{aligned} \mathcal{L}^{\text{dist}} &= \lambda_{l_1} \mathcal{L}_{l_1} + \lambda_{l_1 \text{mask}} \mathcal{L}_{l_1 \text{mask}} \\ &+ \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\text{perc}} \mathcal{L}_{\text{perc}} + \lambda_{\text{style}} \mathcal{L}_{\text{style}} \end{aligned} \quad (1)$$

3.3 Transformation from distance image to line drawing

Next, we describe $G_{\text{edge}} : \mathbb{R}^{H \times W \times 1} \times \{0, 1\}^{H \times W \times 1} \rightarrow \mathbb{R}^{H \times W \times 1}$, a network that converts a distance image $\tilde{V} \in \mathbb{R}^{H \times W \times 1}$ back into a line drawing $\tilde{U} \in \mathbb{R}^{H \times W \times 1}$. Essentially, we can recover the binarized line drawing from the restored distance image analytically. However, because such direct recovery is sensitive to noise, we substituted this operation with a learned CNN. To reduce the number of parameters, we constructed a network with a small number of convolutional layers. Because G_{edge} is a small network, we use the loss function from Equation (1) without the perceptual loss and

Table 1: Comparison by quantitative evaluation

Method	l_1 loss	PSNR	SSIM
EdgeConnect [Nazeri et al. 2019]	0.0155	22.24	0.9428
RFRNet [Li et al. 2020]	0.0151	22.62	0.9452
Ours	0.0138	23.18	0.9504

style loss.

$$\mathcal{L}^{\text{edge}} = \lambda_{l_1} \mathcal{L}_{l_1} + \lambda_{l_1 \text{mask}} \mathcal{L}_{l_1 \text{mask}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} \quad (2)$$

3.4 Inpainting of screentones

Finally, we describe the screentones restoration network $G_{\text{texture}} : \mathbb{R}^{H \times W \times 3} \times \mathbb{R}^{H \times W \times 1} \times \{0, 1\}^{H \times W \times 1} \rightarrow \mathbb{R}^{H \times W \times 3}$. The network takes a missing image I_m , an inpainted line drawing $\tilde{U}$, and a mask M as inputs and returns the restored image $\tilde{I} = G_{\text{texture}}(I_m, \tilde{U}, M) \in \mathbb{R}^{H \times W \times 3}$. The loss function $\mathcal{L}^{\text{texture}}$ is the same as that for G_{dist} with Equation 1.

We use the sum of three losses $\mathcal{L}^{\text{total}} = \mathcal{L}^{\text{dist}} + \mathcal{L}^{\text{edge}} + \mathcal{L}^{\text{texture}}$ as the entire loss function in the training phase. The details of each network are described in the supplementary material.

4 EXPERIMENT

4.1 Dataset

For the dataset, we used the images from the Manga109 dataset [Aizawa et al. 2020; Matsui et al. 2017]. These images were cropped and converted into a uniform size, i.e., 256×256 pixels. Then, we generated the mask corresponding to each image. We describe the details of dataset generation in the supplementary materials. In total, we created 50,000 sets of training data and 10,000 sets of test data.

4.2 Results

We quantitatively compared our proposed method with the existing techniques, EdgeConnect [Nazeri et al. 2019] and RFR-Net [Li et al. 2020]. We trained these methods on the Manga109 dataset. We used

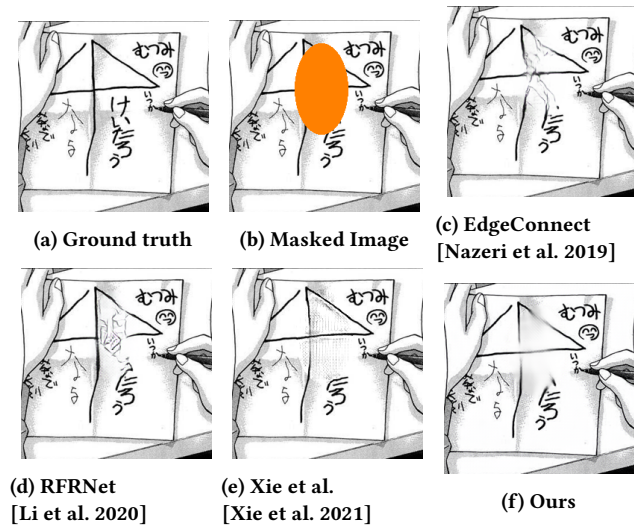


Figure 4: Examples of inpainted image using trained models (LoveHina ©Akamatsu Ken).

the l_1 loss, PSNR, and SSIM as the evaluation measures. Table 1 shows the results. Our proposed method outperformed the existing methods in all evaluation measures.

Figure 4 shows an example of the generated results. Compared with the existing methods, the proposed method clearly recovered the line drawings. In addition to the methods listed in the table, we also compared our method with the method proposed by Xie et al. [Xie et al. 2021]. Both methods are similar because they use distance transform. However, Xie et al. used it only in the training phase to calculate the loss function. We generated the sample images with the model pre-trained by the authors because we could not obtain the training code. The authors have an advantage because they have more training data than what we had. However, we can see that the results of our method contain fewer artifacts than those contained in the results of Xie et al.'s method.

We also inputted a handwritten mask to the trained model instead of an elliptical mask and checked the results. Figure 5 shows the examples of the results. The results show that the objects are erased naturally by completing the lines.

5 CONCLUSION

In this paper, we proposed a method for inpainting comic images. The proposed method converts line drawings into distance images and then recovers the distance images instead of the line drawings. The proposed method can inpaint line drawing images with high-frequency components more accurately. The experiment results showed that the proposed method outperformed the existing methods.

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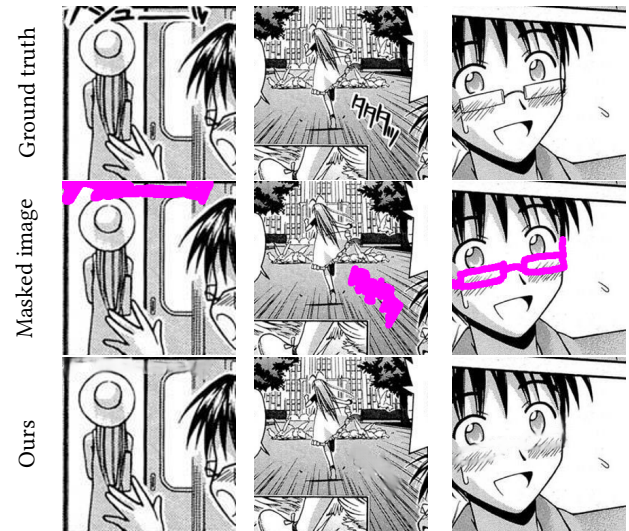


Figure 5: Examples of the generated results when a handwritten mask used as input (LoveHina ©Akamatsu Ken). It can be confirmed that the onomatopoeia has been removed from the input image.

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